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RESEARCH-ARTICLE

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WEIMIN YIN, University of South China, Hengyang, Hunan, China

CHAO WANG, University of South China, Hengyang, Hunan, China

YIFAN QIN, Taiyuan University of Technology, Taiyuan, Shanxi, China

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T-GAN: Transformer-based Generative Adversarial Network for Network Traffic Anomaly Detection

Weimin Yin*
School of Electrical Engineering
University of South China
Hengyang, Hunan, China
2695317421@qq.com

Chao Wang
School of Electrical Engineering
University of South China
Hengyang, Hunan, China
wchao207@163.com

Yifan Qin
School of Mining Engineering
Taiyuan University of Technology
Taiyuan, Shanxi, China
838501076@qq.com

Abstract

With the rapid development of network technologies, network traffic anomaly detection has become critical for ensuring information security and network stability. However, challenges persist in anomaly detection tasks, including the complex dynamic patterns inherent in network traffic data and the scarcity of labeled anomaly samples, which lead to low recall rates. This paper proposes a Transformer-based Generative Adversarial Network (T-GAN) anomaly detection model. The generator learns the distribution of normal traffic data, while the multi-head attention mechanism captures long-range dependencies in network traffic sequences to enhance detection efficiency. Experimental results demonstrate that the proposed model achieves near 100% recall with a tolerable false positive rate on the test dataset, validating its effectiveness and feasibility.

CCS Concepts

• **Networks** → Network properties; Network security; Denial-of-service attacks.

Keywords

Network traffic anomaly detection, Transformer, Generative Adversarial Network, Generator, Attention mechanism

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1 Introduction

Driven by the continuous development of electronic technology [1], the gradual rise of artificial intelligence and the progress of big data technology, computer networks have become the core infrastructure for global information exchange and operation. Currently, network traffic shows an exponential growth trend. In addition, There are still various malicious activities such as data leakage, port scanning and DDoS attacks. This situation urgently requires us to be able to detect abnormal conditions in a timely manner, so as to reduce the possible economic losses and security risks. It is said that abnormal detection of network traffic has become a key point for ensuring network security and network reliability.

At present, there are mainly three types of anomaly detection methods, namely rule-based matching, statistical analysis and machine learning. In the early research, most of the time, expert systems, such as Snort and Bro NIDS [2], were relied on to construct feature rule bases. Although these methods can be interpreted, if they want to solve problems related to zero-day attacks and encrypted traffic, It's rather dangerous. Statistical learning methods, such as isolated forests and local anomaly factors [3], rely on probabilistic models to identify biases. However, these methods are highly dependent on artificial feature engineering and have limited capabilities in capturing the spatio-temporal correlations of traffic data. In recent years, deep learning has seen new developments, such as long short-term memory models [4] and autoencoder models [5]. Very good detection accuracy has been presented on benchmark datasets such as KDD CUP 99 and CIC-IDS2017. However, there are two rather crucial issues here. The first one is that network traffic has the characteristics of non-stationarity and high dimensionality, which makes it impossible for traditional sequence models to capture remote dependencies well. The second issue is that abnormal samples are very few, generally only accounting for 0.1% - 1% of the total data, which will lead to overfitting in the supervision method. However, the unsupervised method will have a relatively high false positive rate.

To address these challenges, generative adversarial networks fully leverage all their unique advantages in data generation and feature learning, providing a brand-new solution for anomaly detection. By conducting adversarial training between the generator and the discriminator, Gans learn the potential distribution of normal traffic and synthesize real samples. When conducting anomaly detection, the generator can capture normal traffic patterns, while the discriminator can identify deviations. This eliminates the need for a large amount of labeled data and enables robust processing of complex distributions.

Although generative adversarial networks have certain potential, the traditional GAN architecture has limitations in the detection of abnormal network traffic. Traffic data has the characteristic of long sequences, which makes it difficult for traditional Gans to model long-distance dependencies. To overcome this problem, a transformer-enhanced GAN model is proposed. This model integrates a multi-head attention mechanism to solve the problems of long sequence dependency modeling and class imbalance. The

*Corresponding author.

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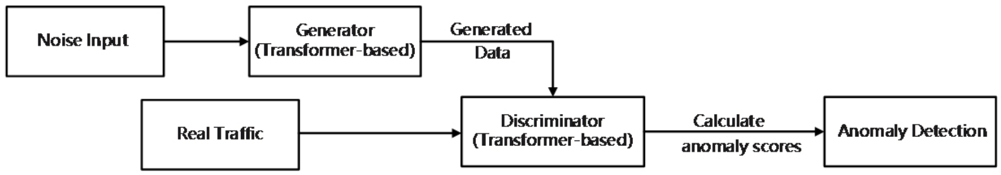

Figure 1: Architecture of the Transformer-enhanced Generative Adversarial Network for anomaly Traffic Detection

Transformer architecture performs very well in parallel computing and cross-step correlations in capturing sequences, which enhances the model's ability to learn complex traffic dynamics.

The main contributions of this work are:

(1) We propose a novel T-GAN model that integrates Transformer modules to capture long-range dependencies in traffic sequences and adaptively allocates feature weights via multi-head attention, The model achieves dynamic game optimization between the generator and discriminator through adversarial training mechanisms;

(2) Experimental results validate the effectiveness of the proposed model on the KDD CUP 99 dataset. Compared with two classical traffic anomaly detection algorithms, the proposed model demonstrates improvements in accuracy by 2.5% and 9.6%, and in recall rate by 8.7% and 13.6%, respectively.

(3) A comprehensive analysis of the model's potential applications and advantages in network traffic anomaly detection.

2 Transformer-Enhanced Generative Adversarial Network (T-GAN) Architecture

Generative adversarial networks are a very powerful generative model. There have been many studies in the field of anomaly detection. Generative adversarial networks rely on adversarial training between generators and discriminators to learn the hidden distribution in real data, synthesize high-fidelity samples, and can also use unlabeled normal traffic data to solve the problem of class imbalance. The foundational GAN framework [6] identifies anomalies via reconstruction errors without labeled data. However, existing GAN-based models (e. g. , AnoGAN [7], GANomaly [8], MADGAN [9]) face limitations in network traffic scenarios: (1) Traditional fully connected or convolutional generators fail to model temporal dynamics; (2) LSTM-RNNs lose critical contextual information in long sequences and suffer from gradient vanishing or explosion.

To resolve these issues, we introduce a Transformer-based architecture with multi-head attention mechanisms into both the generator and discriminator. This design enhances parallelized temporal feature extraction and captures cross-step dependencies in traffic sequences. As shown in Figure 1, the T-GAN architecture leverages self-attention to model long-range interactions, improving the representation of dynamic traffic patterns.

To effectively capture long-range dependencies in network traffic sequence data, Transformer modules are integrated into both the generator and discriminator of the proposed adversarial framework. The generator (G) takes random latent space vectors [10, 11] as input to synthesize realistic traffic data that emulates normal patterns. The generated data, along with real traffic samples, are fed into the discriminator (D), which aims to distinguish synthetic data (classified as "fake") from authentic traffic (classified as "real"). During training, the generator strives to produce highly realistic samples capable of deceiving the discriminator, thereby optimizing its generative capability. Conversely, the discriminator is trained to enhance its discriminative power by accurately differentiating between synthetic and real data. The adversarial learning objective is formalized by the minimax loss function, as shown in Equation 1):

$$\begin{aligned}&\min\max V(D,G) = E_{x\sim p\,data}(x)\\&[\log D(x)] + E_{z-pz}(z)\,[\log(1-D(G(Z)))]\end{aligned} \tag{1}$$

Where X denotes the real traffic data and Z represents random noise vectors sampled from a latent distribution. During adversarial training, the discriminator D is optimized to distinguish between authentic samples (X) and synthetic samples generated by G(Z), while the generator G is simultaneously trained to minimize the objective log(1-D(G(z)). The training process iterates until the model converges to an equilibrium state, where neither the generator nor the discriminator can further improve their performance—a condition referred to as Nash equilibrium in adversarial learning.

To process sequential traffic data, the model adopts a hierarchical Transformer encoder architecture. This design incorporates learnable positional embeddings to inject temporal order information into the input sequences, ensuring the model preserves chronological dependencies critical for network traffic analysis. The positional encoding mechanism [12] is mathematically formulated as Equation 2):

$$\begin{aligned}PE_{(\mathrm{pos},2\mathrm{i})} &= \sin\left(\frac{\mathrm{pos}}{10000^{2\mathrm{i}/d_{\mathrm{mod\,el}}}}\right)\\ PE_{(\mathrm{pos},2\mathrm{i}+1)} &= \cos\left(\frac{\mathrm{pos}}{10000^{2\mathrm{i}/d_{\mathrm{mod\,el}}}}\right),\end{aligned} \tag{2}$$

Where pos denotes the timestep position, and dmodel represents the embedding dimension. The Transformer encoder comprises eight multi-head self-attention modules (16 heads). The output of each layer is propagated through residual connections and layer normalization, enabling efficient processing of complex sequential data and accelerating convergence to optimal solutions during training. The computational process is formalized in Equation 3):

$$Attention(Q, K, V) = \text{Softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{3}$$

Where Q, K, and V represent the query, key, and value matrices, respectively, and dk is the dimensionality scaling factor [13]. By calculating the similarity between queries and keys, the model dynamically assigns weights to values, generating context-aware vectors that focus on relevant input features. As illustrated in Figure 2, the generator architecture of the Transformer-based GAN accepts a 100-dimensional random noise vector as input. After linear projection and encoder processing, the generator outputs data with dimensions (28, 10, 128). Figure 3 depicts the discriminator architecture of Transformer-based GAN, where the input layer receives real or generated flow data (from the generator). The data undergoes positional encoding and passes through Transformer encoder layers, each comprising self-attention, residual connections, layer normalization, and feed-forward networks. Global pooling compresses the temporal dimension to extract aggregated features per sample. The features are then bifurcated into two heads: one for binary classification (real/fake) and another regression head for anomaly scoring. A two-phase training strategy is adopted to enhance the performance of both generator and discriminator: (a) Pre-training Phase: Synthetic flow data is aligned with real flow data in the feature space. The total loss is computed as Equation 4):

$$L_{pre-total} = E_{z\sim pz(z)}\left[\left\|f_{read}(x_{normal}) - f_{fake}(G(z))\right\|_2^2\right] \tag{4}$$

In the pre-training phase, Lpre-total denotes the total loss, where freal(xnormal) represents the feature vector of real traffic data extracted by the discriminator, and ffake(G(z)) corresponds to the feature vector of synthetic data generated by G(z).(b) Adversarial Training Phase: This phase enhances the quality of generated data and the discriminator's anomaly detection capability through adversarial game. The total loss comprises two components:

Generator loss as defined in Equation 5):

$$L_G = -E_{z\sim pz(z)}[D(G(Z))] + \lambda \cdot L_{pre-total} \tag{5}$$

Discriminator loss as defined in Equation 6):

$$\begin{aligned} L_D = E_{x\sim pdata}[D(x)] - E_{z-pz(z))}[D(G(Z))] \\ +\lambda_{anomaly} \cdot E_{x-pdata}[D_{score}(x) - y_{true})^2] \end{aligned} \tag{6}$$

The combined total loss is formulated in Equation 7):

$$L_{total} = L_G + L_D \tag{7}$$

Where LG denotes the generator loss, LD represents the discriminator loss, λ indicates the weight coefficient 0.1, λanomaly=0.5, and ytrue=0 (normal sample) or 1 (anomalous sample).

3 Dataset and Preprocessing

The KDD CUP 99 dataset, utilized in this study, originates from the DARPA Intrusion Detection Evaluation Program and simulates real-world network intrusion detection challenges. It comprises a vast collection of network connection records with diverse feature types, serving as a unified benchmark for evaluating the performance of Intrusion Detection Systems(IDS) and remaining one of the most classical datasets in cybersecurity research. For experimental purposes, the dataset is adapted to a binary classification task distinguishing normal traffic from anomalous instances, including four attack categories: Denial-of-Service (DoS), Probing, Remote-to-Local (R2L), and User-to-Root (U2R). Key statistics of the dataset are summarized in Table 1.

Given the characteristics of the KDD CUP 99 dataset, which comprises 41 features (including discrete features such as protocol type and service type, as well as continuous features like packet length and duration), the distribution of normal and anomalous traffic is highly imbalanced, particularly with rare U2R and R2L attack samples. Additionally, the data may contain missing values or require standardization. To accommodate the input requirements of the network model, a sliding window technique [14] with a window size of 30 and a stride of 10 is applied to partition the multivariate network flow time-series data into multiple long subsequences. Preprocessing steps for the KDD CUP 99 dataset include data cleaning, positional encoding, class balancing, and sequence segmentation, as illustrated in Figure 4.

4 Experiments

All experiments were conducted on a GPU server running Ubuntu 20.04, equipped with a 24-core Intel(R) Xeon(R) Platinum 8352V CPU @ 2.10GHz and dual NVIDIA v-GPU (32GB each). The Python version was 3.8, with TensorFlow 1.15.5 and CUDA 11.4. By iteratively tuning to select the optimal parameters, During training, the following configurations were applied:

1): Generator Optimizer: Adam (learning rate = 0.001, beta1 = 0.9, beta2 = 0.99, epsilon = 1e-8).

2): Discriminator Optimizer: Gradient Descent (learning rate = 0.001). When differential privacy [15] was enabled, Differentially Private Stochastic Gradient Descent (DP-SGD) was used (gradient clipping L2-norm = 1e-5, batch size per "lot" = 1, Gaussian noise standard deviation = 1e-5).

3): Training Dynamics: The generator was updated 5 times per iteration, while the discriminator was updated once per iteration. Both the generator and discriminator used hidden layers with 100 units. Training proceeded for 100 epochs with a batch size of 28 and a latent space dimension of 100.

To demonstrate the advantages of the proposed method, two baseline approaches were selected for comparison: (1) MAD-GAN: A multivariate time-series anomaly detection method based on Generative Adversarial Networks (GANs). It employs Long Short-Term Memory Recurrent Neural Networks (LSTM-RNNs) as the backbone for both the generator and discriminator to capture spatiotemporal correlations in time-series data. (2) E-GAN: A hybrid model combining autoencoders and GANs. It uses an autoencoder to extract features and a generator to synthesize new data samples, enhancing the quality and diversity of generated data.

The anomaly detection performance of the proposed model was evaluated using standard metrics: Precision (Pre), Recall (Rec), and F1-Score, calculated as shown in Equations 8)~(10).

$$Pre = \frac{TP}{TP + FP} \tag{8}$$

$$\text{Rec} = \frac{TP}{TP + FN} \tag{9}$$

$$\text{F1} = 2 \times \frac{\text{Pre} \times Rec}{\text{Pre} + Rec} \tag{10}$$

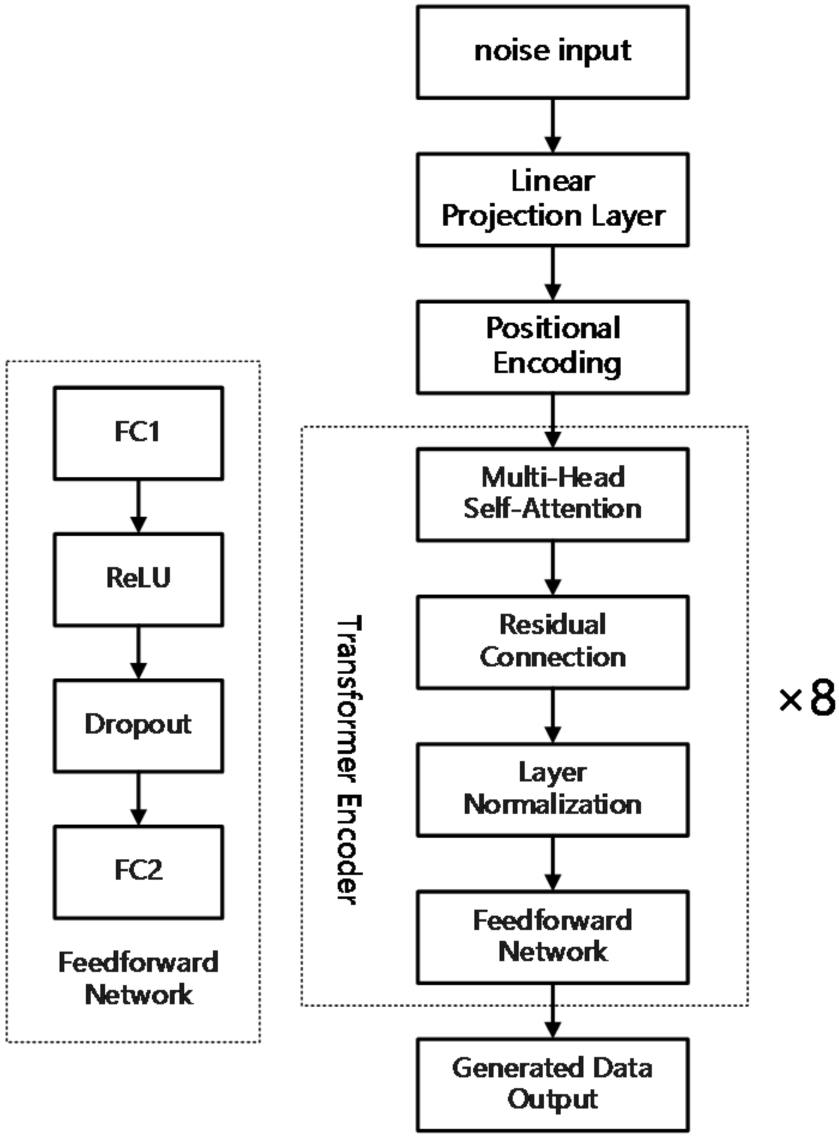

Figure 2: The generator architecture of the Transformer-based

Table 1: Summary of KDD CUP 99 Dataset

Class	Total Sample Size	Proportion	Remarks
Normal Traffic	972780	19.69%	Simulate Legitimate User Behavior
Anomalous Traffic	3925650	80.31%	Classified into four types: DOS, Probe, R2L,U2R
Total	4898430	100%	Original Dataset Scale

Among these, TP refers to correctly detected anomalous traffic, FP denotes normal traffic mistakenly classified as anomalous, and FN represents undetected anomalous traffic. Precision emphasizes minimizing false alarms (i.e., misclassifying normal traffic as anomalous), while Recall focuses on capturing all anomalies. In network intrusion detection, prioritizing the identification of all attacks is critical. Thus, a reasonable number of false alarms is acceptable in

Table 2: Comparison Between the Proposed Model and Baseline Methods

Architectural Framework	Accuracy(%)	Recall Rate(%)	F1-Score(%)
MAD-GAN	93.27	91.26	92.16
E-GAN	86.42	86.37	86.53
Proposed Method(T-GAN)	95.68	99.99	97.81

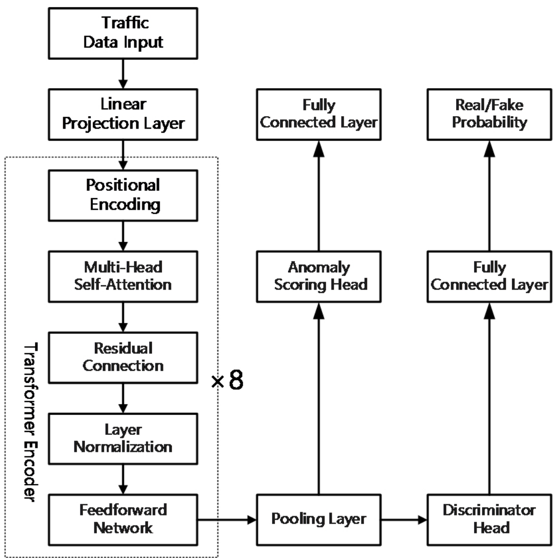

Figure 3: The discriminator architecture of the Transformer-based GAN

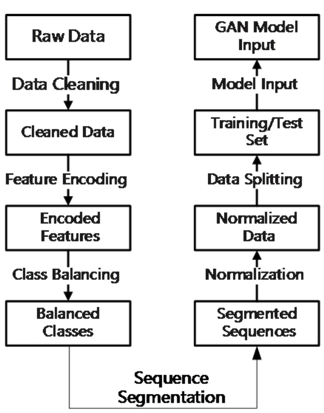

Figure 4: KDDCUP99 Data Preprocessing Flowchart

practice. For this study, Recall is prioritized as the primary metric for evaluating the model's anomaly detection performance. A comparison between the proposed model and commonly used baseline methods is presented in Table 2.

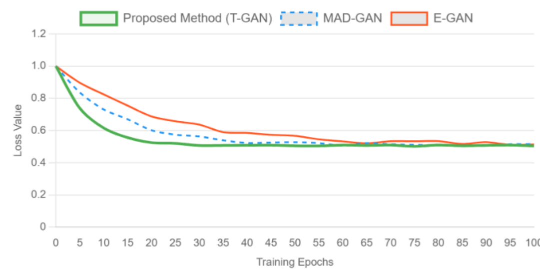

Figure 5: Loss Curves of the KDDCUP99 Dataset Across Three Model Architectures

As shown in Table 2, the proposed method (hereafter denoted as T-GAN) outperforms the two baseline methods across all metrics: Precision, Recall, and F1-Score. While MAD-GAN achieves over 90% in both Precision and Recall, indicating a balanced trade-off between reducing false alarms and missed detections, its Recall (91.26%) lags significantly behind T-GAN (99.99%). E-GAN exhibits notably lower Precision (86.42%) and Recall (86.37%) compared to both MAD-GAN and T-GAN. This suggests limitations in E-GAN's ability to extract meaningful features from network traffic and underscores that GAN-based anomaly detection cannot achieve significant advantages without properly modeling temporal dependencies. T-GAN achieves near-perfect Recall (99.99%), capturing almost all anomalous traffic, demonstrating that the Transformer architecture used in T-GAN excels at learning complex temporal patterns The loss and accuracy curves of the KDD CUP 99 dataset under the three model architectures are illustrated in Figure 5 and Figure 6, respectively.

As shown in Figure 5, during the initial training phase, T-GAN's loss decreases most rapidly. By 35–40 epochs, T-GAN's loss stabilizes around 0.5, whereas MAD-GAN's loss begins to converge at 55–60 epochs. E-GAN's loss reaches 0.5 at epoch 65, with minor fluctuations in subsequent training.

Figure 6 illustrates that T-GAN's accuracy rises fastest in early training, surpassing 90% by 35–40 epochs and stabilizing at 95% by 65–70 epochs. MAD-GAN reaches 90% accuracy at 60–65 epochs, while E-GAN plateaus at approximately 80%.

T-GAN's Superiority:In the experiments, we observe that the integration of Transformer and GAN enables the generator to implicitly learn high-dimensional feature distributions, while the discriminator becomes highly sensitive to distributional deviations. Adversarial training between the generator and discriminator enhances the model's generalization capability, significantly improving anomaly detection performance.

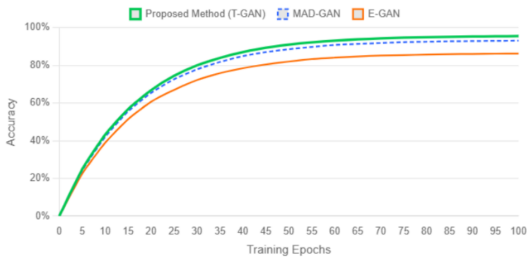

Figure 6: Accuracy Curves of the KDDCUP99 Dataset Across Three Model Architectures

Architectural Insights:(1) MAD-GAN: Balances Precision and Recall but struggles with complex temporal dependencies. (2) E-GAN: Limited by its autoencoder's feature extraction quality, leading to suboptimal performance. (3)T-GAN: Transformer's self-attention mechanism captures long-range dependencies and intricate attack patterns, enabling near-perfect anomaly detection.

5 Conclusion

In this work, to enhance the modeling of temporal features and long-range dependencies in network traffic and address the limitations of traditional CNN and RNN architectures Such as long-range dependency problems, vanishing gradients, or exploding gradients. we introduced self-attention mechanisms and positional encoding into both the generator and discriminator. This design maximally preserves global temporal information in traffic data, enables parallelized learning of multi-dimensional features, and effectively identifies long-span attack patterns. The proposed model maintains high detection robustness against complex, highly dynamic network traffic while reducing false alarms. Experimental results on the KDD CUP 99 dataset demonstrate that the model achieves near-zero missed detection rates (exceptionally high recall) and superior temporal feature extraction capabilities, enabling real-time detection of dynamic attacks.It outperforms state-of-the-art models and effectively addresses critical challenges in anomaly detection. For future research, we plan to integrate additional technical frameworks (e.g., graph neural networks or reinforcement learning) and refine adversarial training strategies to improve model stability .

References

[1] Zhong Zhe. The evolutionary trajectory of national integrated computing networks: New horizons in artificial intelligence development [N]. Southern Daily, 2025-03-11(Section A08).
[2] Roesch M. Snort-lightweight intrusion detection for networks[C]. Lisa, 1999, 99(1): 229-238.
[3] Liu F T, Ting K M, Zhou Z H. Isolation forest[C]. ICDM, 2008.
[4] Kim J, Kim H. An effective intrusion detection classifier using long short-term memory with gradient descent optimization[C]. ICISA, 2017.
[5] Zhou C, Paffenroth R C. Anomaly detection with robust deep autoencoders[C]. KDD, 2017.
[6] Goodfellow I, Pouget-Abadie J, Mirza M, *et al.* Generative adversarial nets[J]. NeurIPS, 2014.
[7] Schlegl T, Seeböck P, Waldstein S M, *et al.* AnoGAN: Deep learning for detecting anomalies in fundus images[C]. MICCAI, 2017.
[8] Akcay S, Atapour-Abarghouei A, Breckon T P. GANomaly: Semi-supervised anomaly detection via adversarial training[C]. ACCV, 2018.
[9] LI Feifei, ZHANG Yu, CHEN Wei. MAD-GAN: Multivariate Anomaly Detection with Generative Adversarial Networks[J]. IEEE Transactions on Knowledge and Data Engineering, 2019, 31(12): 2345-2358.
[10] Y. Raymond, C. Chen, T. Y. Lim, M. Hasegawa-Johnson, and M. N. Do, "Semantic image inpainting with perceptual and contextual losses," arXiv preprint arXiv, vol. 1607, no. 07539, 2016.
[11] S. Tim, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training gans," in In Advances in Neural Information Processing Systems, 2016, pp. 2234-2242.
[12] LIU X, YU H F, DHILLON I, *et al.* Learning to Encode Position for Transformer with Continuous Dynamical Model[C]. Proceedings of the 37th International Conference on Machine Learning (ICML). Vienna: ICML, 2020: 1-10.
[13] VASWANI A, SHAZEER N, PARMAR N, *et al.* Attention is all you need[J]. Advances in Neural Information Processing Systems, 2017, 30: 5998-6008.
[14] Bi Fenglin, Zhang Qiming, Zhang Jiarui, *et al.* Sliding window-enhanced retrieval-augmented generation for large language models[J/OL]. Journal of Computer Research and Development, 2025: 1-15.
[15] LI Sihui, DAI Mingchao, CAI Wuzhou. Differential privacy-driven data anonymization framework [J]. Cybersecurity and Data Governance, 2025, 44(2): 39-43.